

Review

Deep learning in agriculture: A survey

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ABSTRACT

Deep learning constitutes a recent, modern technique for image processing and data analysis, with promising results and large potential. As deep learning has been successfully applied in various domains, it has recently entered also the domain of agriculture. In this paper, we perform a survey of 40 research efforts that employ deep learning techniques, applied to various agricultural and food production challenges. We examine the particular agricultural problems under study, the specific models and frameworks employed, the sources, nature and pre-processing of data used, and the overall performance achieved according to the metrics used at each work under study. Moreover, we study comparisons of deep learning with other existing popular techniques, in respect to differences in classification or regression performance. Our findings indicate that deep learning provides high accuracy, outperforming existing commonly used image processing techniques.

1. Introduction

Smart farming (Tyagi, 2016) is important for tackling the challenges of agricultural production in terms of productivity, environmental impact, food security and sustainability (Gebbers and Adamchuk, 2010). As the global population has been continuously increasing (Kitzes et al., 2008), a large increase on food production must be achieved (FAO, 2009), maintaining at the same time availability and high nutritional quality across the globe, protecting the natural ecosystems by using sustainable farming procedures.

To address these challenges, the complex, multivariate and unpredictable agricultural ecosystems need to be better understood by monitoring, measuring and analyzing continuously various physical aspects and phenomena. This implies analysis of big agricultural data (Kamilaris et al., 2017b), and the use of new information and communication technologies (ICT) (Kamilaris et al., 2016), both for short-scale crop/farm management as well as for larger-scale ecosystems' observation, enhancing the existing tasks of management and decision/policy making by context, situation and location awareness. Larger-scale observation is facilitated by remote sensing (Bastiaanssen et al., 2000), performed by means of satellites, airplanes and unmanned aerial vehicles (UAV) (i.e. drones), providing wide-view snapshots of the agricultural environments. It has several advantages when applied to agriculture, being a well-known, non-destructive method to collect information about earth features while data may be obtained systematically over large geographical areas.

A large subset of the volume of data collected through remote

sensing involve images. Images constitute, in many cases, a complete picture of the agricultural environments and could address a variety of challenges (Liaghat and Balasundram, 2010; Ozdogan et al., 2010). Hence, imaging analysis is an important research area in the agricultural domain and intelligent data analysis techniques are being used for image identification/classification, anomaly detection etc., in various agricultural applications (Teke et al., 2013; Saxena and Armstrong, 2014; Singh et al., 2016). The most popular techniques and applications are presented in Appendix A, together with the sensing methods employed to acquire the images. From existing sensing methods, the most common one is satellite-based, using multi-spectral and hyperspectral imaging. Synthetic aperture radar (SAR), thermal and near infrared (NIR) cameras are being used in a lesser but increasing extent (Ishimwe et al., 2014), while optical and X-ray imaging are being applied in fruit and packaged food grading. The most popular techniques used for analyzing images include machine learning (ML) (K-means, support vector machines (SVM), artificial neural networks (ANN) amongst others), linear polarizations, wavelet-based filtering, vegetation indices (NDVI) and regression analysis (Saxena and Armstrong, 2014; Singh et al., 2016).

Besides the aforementioned techniques, a new one which is recently gaining momentum is deep learning (DL) (LeCun et al., 2015; LeCun and Bengio, 1995). DL belongs to the machine learning computational field and is similar to ANN. However, DL is about “deeper” neural networks that provide a hierarchical representation of the data by means of various convolutions. This allows larger learning capabilities and thus higher performance and precision. A brief description of DL is

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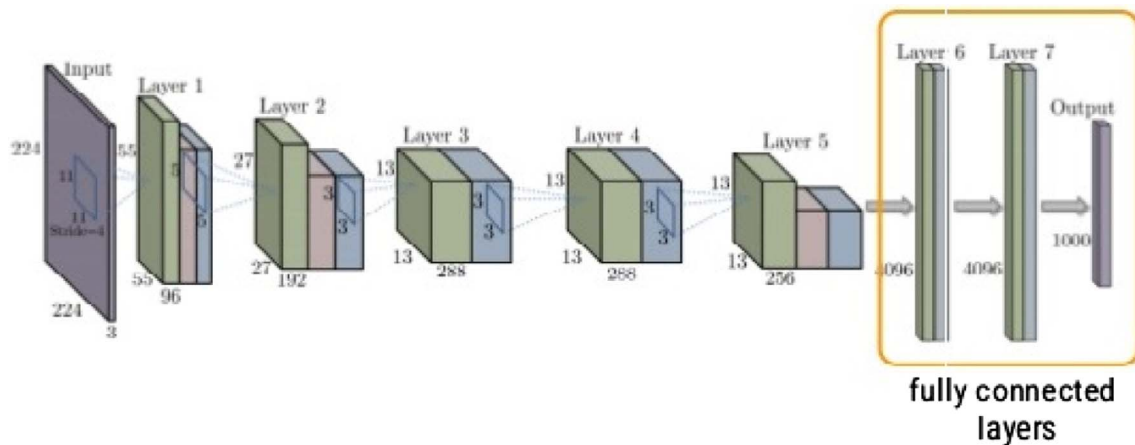


Fig. 1. CaffeNet, an example CNN architecture.

Source: Sladojevic et al. (2016).

attempted in Section 3.

The motivation for preparing this survey stems from the fact that DL in agriculture is a recent, modern and promising technique with growing popularity, while advancements and applications of DL in other domains indicate its large potential. The fact that today there exists at least 40 research efforts employing DL to address various agricultural problems with very good results, encouraged the authors to prepare this survey. To the authors' knowledge, this is the first such survey in the agricultural domain, while a small number of more general surveys do exist (Deng and Yu, 2014; Wan et al., 2014; Najafabadi et al., 2015), covering related work in DL in other domains.

2. Methodology

The bibliographic analysis in the domain under study involved two steps: (a) collection of related work and (b) detailed review and analysis of this work. In the first step, a keyword-based search for conference papers or journal articles was performed from the scientific databases IEEE Xplore and ScienceDirect, and from the web scientific indexing services Web of Science and Google Scholar. As search keywords, we used the following query:

["deep learning"] AND ["agriculture" OR "farming"]

In this way, we filtered out papers referring to DL but not applied to the agricultural domain. From this effort, 47 papers had been initially identified. Restricting the search for papers with appropriate application of the DL technique and meaningful findings¹, the initial number of papers reduced to 40.

In the second step, the 40 papers selected from the previous step were analyzed one-by-one, considering the following research questions:

1. Which was the agricultural- or food-related problem they addressed?
2. Which was the general approach and type of DL-based models employed?
3. Which sources and types of data had been used?
4. Which were the classes and labels as modeled by the authors? Were there any variations among them, observed by the authors?
5. Any pre-processing of the data or data augmentation techniques used?
6. Which has been the overall performance depending on the metric

adopted?

7. Did the authors test the performance of their models on different datasets?
8. Did the authors compare their approach with other techniques and, if yes, which was the difference in performance?

Our main findings are presented in Section 4 and the detailed information per paper is listed in Appendix B.

3. Deep learning

DL extends classical ML by adding more "depth" (complexity) into the model as well as transforming the data using various functions that allow data representation in a hierarchical way, through several levels of abstraction (Schmidhuber, 2015; LeCun and Bengio, 1995). A strong advantage of DL is feature learning, i.e. the automatic feature extraction from raw data, with features from higher levels of the hierarchy being formed by the composition of lower level features (LeCun et al., 2015). DL can solve more complex problems particularly well and fast, because of more complex models used, which allow massive parallelization (Pan and Yang, 2010). These complex models employed in DL can increase classification accuracy or reduce error in regression problems, provided there are adequately large datasets available describing the problem. DL consists of various different components (e.g. convolutions, pooling layers, fully connected layers, gates, memory cells, activation functions, encode/decode schemes etc.), depending on the network architecture used (i.e. Unsupervised Pre-trained Networks, Convolutional Neural Networks, Recurrent Neural Networks, Recursive Neural Networks).

The highly hierarchical structure and large learning capacity of DL models allow them to perform classification and predictions particularly well, being flexible and adaptable for a wide variety of highly complex (from a data analysis perspective) challenges (Pan and Yang, 2010). Although DL has met popularity in numerous applications dealing with raster-based data (e.g. video, images), it can be applied to any form of data, such as audio, speech, and natural language, or more generally to continuous or point data such as weather data (Sehgal et al., 2017), soil chemistry (Song et al., 2016) and population data (Demmers et al., 2012). An example DL architecture is displayed in Fig. 1, illustrating CaffeNet (Jia et al., 2014), an example of a convolutional neural network, combining convolutional and fully connected (dense) layers.

Convolutional Neural Networks (CNN) constitute a class of deep, feed-forward ANN, and they appear in numerous of the surveyed papers as the technique used (17 papers, 42%). As the figure shows, various convolutions are performed at some layers of the network, creating different representations of the learning dataset, starting from more

¹ A small number of papers claimed of using DL in some agricultural-related application, but they did not show any results nor provided performance metrics that could indicate the success of the technique used.

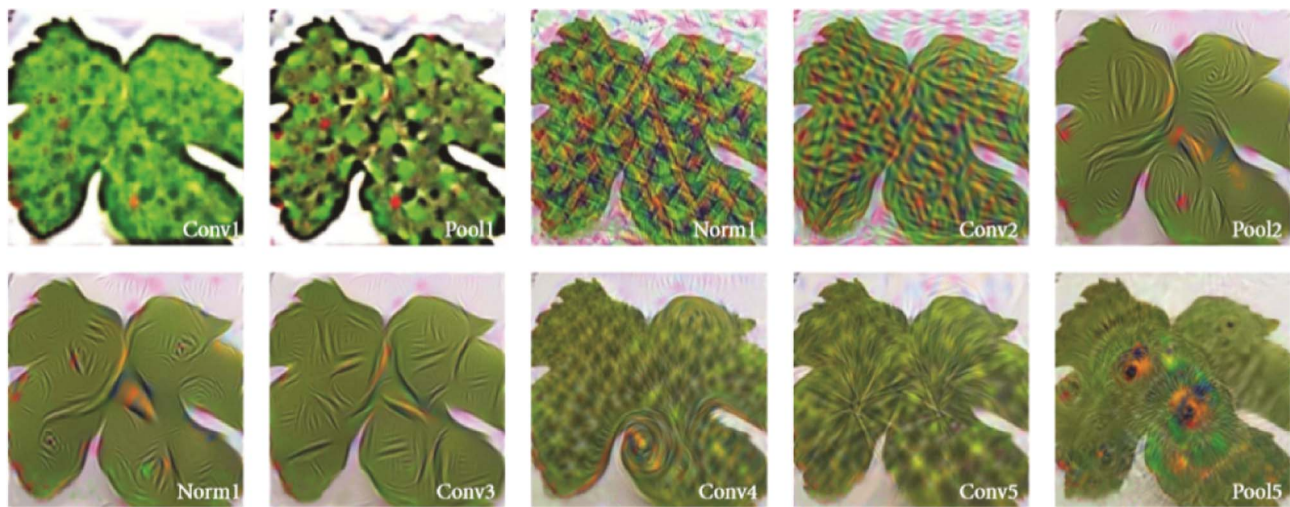


Fig. 2. Visualization of the output layers images after each processing step of the CaffeNet CNN (i.e. convolution, pooling, normalization) at a plant disease identification problem based on leaf images.

Source: Sladojevic et al. (2016).

general ones at the first larger layers, becoming more specific at the deeper layers. The convolutional layers act as feature extractors from the input images whose dimensionality is then reduced by the pooling layers. The convolutional layers encode multiple lower-level features into more discriminative features, in a way that is spatially context-aware. They may be understood as banks of filters that transform an input image into another, highlighting specific patterns. The fully connected layers, placed in many cases near the output of the model, act as classifiers exploiting the high-level features learned to classify input images in predefined classes or to make numerical predictions. They take a vector as input and produce another vector as output. An example visualization of leaf images after each processing step of the CaffeNet CNN, at a problem of identifying plant diseases, is depicted in Fig. 2. We can observe that after each processing step, the particular elements of the image that reveal the indication of a disease become more evident, especially at the final step (Pool5).

One of the most important advantages of using DL in image processing is the reduced need of feature engineering (FE). Previously, traditional approaches for image classification tasks had been based on hand-engineered features, whose performance affected heavily the overall results. FE is a complex, time-consuming process which needs to be altered whenever the problem or the dataset changes. Thus, FE constitutes an expensive effort that depends on experts' knowledge and does not generalize well (Amara et al., 2017). On the other hand, DL does not require FE, locating the important features itself through training.

A disadvantage of DL is the generally longer training time. However, testing time is generally faster than other methods ML-based methods (Chen et al., 2014). Other disadvantages include problems that might occur when using pre-trained models on datasets that are small or significantly different, optimization issues because of the models' complexity, as well as hardware restrictions.

In Section 5, we discuss over advantages and disadvantages of DL as they reveal through the surveyed papers.

3.1. Available architectures, datasets and tools

There exist various successful and popular architectures, which researchers may use to start building their models instead of starting from scratch. These include AlexNet (Krizhevsky et al., 2012), CaffeNet (Jia et al., 2014) (displayed in Fig. 1), VGG (Simonyan and Zisserman, 2014), GoogleNet (Szegedy et al., 2015) and Inception-ResNet (Szegedy et al., 2017), among others. Each architecture has different advantages

and scenarios where it is more appropriate to be used (Canziani et al., 2016). It is also worth noting that almost all of the aforementioned models come along with their weights pre-trained, which means that their network had been already trained by some dataset and has thus learned to provide accurate classification for some particular problem domain (Pan and Yang, 2010). Common datasets used for pre-training DL architectures include ImageNet (Deng et al., 2009) and PASCAL VOC (PASCAL VOC Project, 2012) (see also Appendix C).

Moreover, there exist various tools and platforms allowing researchers to experiment with DL (Bahrampour et al., 2015). The most popular ones are Theano, TensorFlow, Keras (which is an application programmer's interface on top of Theano and TensorFlow), Caffe, PyTorch, TFLearn, Pylearn2 and the Deep Learning Matlab Toolbox. Some of these tools (i.e. Theano, Caffe) incorporate popular architectures such as the ones mentioned above (i.e. AlexNet, VGG, GoogleNet), either as libraries or classes. For a more elaborate description of the DL concept and its applications, the reader could refer to existing bibliography (Schmidhuber, 2015; Deng and Yu, 2014; Wan et al., 2014; Najafabadi et al., 2015; Canziani et al., 2016; Bahrampour et al., 2015).

4. Deep learning applications in agriculture

In Appendix B, we list the 40 identified relevant works, indicating the agricultural-related research area, the particular problem they address, DL models and architectures implemented, sources of data used, classes and labels of the data, data pre-processing and/or augmentation employed, overall performance achieved according to the metrics adopted, as well as comparisons with other techniques, wherever available.

4.1. Areas of use

Sixteen areas have been identified in total, with the popular ones being identification of weeds (5 papers), land cover classification (4 papers), plant recognition (4 papers), fruits counting (4 papers) and crop type classification (4 papers).

It is remarkable that all papers, except from Demmers et al. (2010, 2012) and Chen et al. (2014), were published during or after 2015, indicating how recent and modern this technique is, in the domain of agriculture. More precisely, from the remaining 37 papers, 15 papers have been published in 2017, 15 in 2016 and 7 in 2015.

The large majority of the papers deal with image classification and identification of areas of interest, including detection of obstacles (e.g.

(Steen et al., 2016; Christiansen et al., 2016)) and fruit counting (e.g. (Rahneemoonfar and Sheppard, 2017; Sa et al., 2016)). Some papers focus on predicting future parameters, such as corn yield (Kuwata and Shibasaki, 2015) soil moisture content at the field (Song et al., 2016) and weather conditions (Sehgal et al., 2017).

From another perspective, most papers (20) target crops, while few works consider issues such as weed detection (8 papers), land cover (4 papers), research on soil (2 papers), livestock agriculture (3 papers), obstacle detection (3 papers) and weather prediction (1 paper).

4.2. Data sources

Observing the sources of data used to train the DL model at every paper, large datasets of images are mainly used, containing thousands of images in some cases, either real ones (e.g. (Mohanty et al., 2016; Reyes et al., 2015; Dyrmann et al., 2016a)), or synthetic produced by the authors (Rahneemoonfar and Sheppard, 2017; Dyrmann et al., 2016b). Some datasets originate from well-known and publicly-available datasets such as PlantVillage, LifeCLEF, MalayaKew, UC Merced and Flavia (see Appendix C), while others constitute sets of real images collected by the authors for their research needs (e.g. (Sladojevic et al., 2016; Bargoti and Underwood, 2016; Xinshao and Cheng, 2015; Sørensen et al., 2017)). Papers dealing with land cover, crop type classification and yield estimation, as well as some papers related to weed detection employ a smaller number of images (e.g. tens of images), produced by UAV (Lu et al., 2017; Rebetez et al., 2016; Milioto et al., 2017), airborne (Chen et al., 2014; Luus et al., 2015) or satellite-based remote sensing (Kussul et al., 2017; Minh et al., 2017; Ienco et al., 2017; Rußwurm and Körner, 2017). A particular paper investigating segmentation of root and soil uses images from X-ray tomography (Douarre et al., 2016). Moreover, some papers use text data, collected either from repositories (Kuwata and Shibasaki, 2015; Sehgal et al., 2017) or field sensors (Song et al., 2016; Demmers et al., 2010, 2012). In general, the more complicated the problem to be solved, the more data is required. For example, problems involving large number of classes to identify (Mohanty et al., 2016; Reyes et al., 2015; Xinshao and Cheng, 2015) and/or small variation among the classes (Luus et al., 2015; Rußwurm and Körner, 2017; Yalcin, 2017; Namin et al., 2017; Xinshao and Cheng, 2015), require large number of input images to train their models.

4.3. Data variation

Variation between classes is necessary for the DL models to be able to differentiate features and characteristics, and perform accurate classifications.² Hence, accuracy is positively correlated with variation among classes. Nineteen papers (47%) revealed some aspects of poor data variation. Luus et al. (2015) observed high relevance between some land cover classes (i.e. medium density and dense residential, buildings and storage tanks) while Ienco et al. (2017) found that tree crops, summer crops and truck farming were classes highly mixed. A confusion between maize and soybeans was evident in Kussul et al. (2017) and variation was low in botanically related crops, such as meadow, fallow, triticale, wheat, and rye (Rußwurm and Körner, 2017). Moreover, some particular views of the plants (i.e. flowers and leaf scans) offer different classification accuracy than branches, stems and photos of the entire plant. A serious issue in plant phenology recognition is the fact that appearances change very gradually and it is challenging to distinguish images falling into the growing durations that are in the middle of two successive stages (Yalcin, 2017; Namin et al., 2017). A similar issue appears when assessing the quality of vegetative development (Minh et al., 2017). Furthermore, in the challenging problem of fruit counting, the models suffer from high

occlusion, depth variation, and uncontrolled illumination, including high color similarity between fruit/foilage (Chen et al., 2017; Bargoti and Underwood, 2016).

Finally, identification of weeds faces issues with respect to lighting, resolution, and soil type, and small variation between weeds and crops in shape, texture, color and position (i.e. overlapping) (Dyrmann et al., 2016a; Xinshao and Cheng, 2015; Dyrmann et al., 2017). In the large majority of the papers mentioned above (except from Minh et al. (2017)), this low variation has affected classification accuracy significantly, i.e. more than 5%.

4.4. Data pre-processing

The large majority of related work (36 papers, 90%) involved some image pre-processing steps, before the image or particular characteristics/features/statistics of the image were fed as an input to the DL model. The most common pre-processing procedure was image resize (16 papers), in most cases to a smaller size, in order to adapt to the requirements of the DL model. Sizes of 256×256 , 128×128 , 96×96 and 60×60 pixels were common. Image segmentation was also a popular practice (12 papers), either to increase the size of the dataset (Ienco et al., 2017; Rebetez et al., 2016; Yalcin, 2017) or to facilitate the learning process by highlighting regions of interest (Sladojevic et al., 2016; Mohanty et al., 2016; Grinblat et al., 2016; Sa et al., 2016; Dyrmann et al., 2016a; Potena et al., 2016) or to enable easier data annotation by experts and volunteers (Chen et al., 2017; Bargoti and Underwood, 2016). Background removal (Mohanty et al., 2016; McCool et al., 2017; Milioto et al., 2017), foreground pixel extraction (Lee et al., 2015) or non-green pixels removal based on NDVI masks (Dyrmann et al., 2016a; Potena et al., 2016) were also performed to reduce the datasets' overall noise.

Other operations involved the creation of bounding boxes (Chen et al., 2017; Sa et al., 2016; McCool et al., 2017; Milioto et al., 2017) to facilitate detection of weeds or counting of fruits. Some datasets were converted to grayscale (Santoni et al., 2015; Amara et al., 2017) or to the HSV color model (Luus et al., 2015; Lee et al., 2015).

Furthermore, some papers used features extracted from the images as input to their models, such as shape and statistical features (Hall et al., 2015), histograms (Hall et al., 2015; Xinshao and Cheng, 2015; Rebetez et al., 2016), Principal Component Analysis (PCA) filters (Xinshao and Cheng, 2015), Wavelet transformations (Kuwata and Shibasaki, 2015) and Gray Level Co-occurrence Matrix (GLCM) features (Santoni et al., 2015). Satellite or aerial images involved a combination of pre-processing steps such as orthorectification (Lu et al., 2017; Minh et al., 2017) calibration and terrain correction (Kussul et al., 2017; Minh et al., 2017) and atmospheric correction (Rußwurm and Körner, 2017).

4.5. Data augmentation

It is worth-mentioning that some of the related work under study (15 papers, 37%) employed data augmentation techniques (Krizhevsky et al., 2012), to enlarge artificially their number of training images. This helps to improve the overall learning procedure and performance, and for generalization purposes, by means of feeding the model with varied data. This augmentation process is important for papers that possess only small datasets to train their DL models, such as (Bargoti and Underwood, 2016; Sladojevic et al., 2016; Sørensen et al., 2017; Mortensen et al., 2016; Namin et al., 2017 and Chen et al., 2017). This process was especially important in papers where the authors trained their models using synthetic images and tested them on real ones (Rahneemoonfar and Sheppard, 2017 and Dyrmann et al., 2016b). In this case, data augmentation allowed their models to generalize and be able to adapt to the real-world problems more easily.

Transformations are label-preserving, and included rotations (12 papers), dataset partitioning/cropping (3 papers), scaling (3 papers),

² Classification accuracy is defined in Section 4.7 and Table 1.

transposing (Sørensen et al., 2017), mirroring (Dyrmann et al., 2016a), translations and perspective transform (Sladojevic et al., 2016), adaptations of objects' intensity in an object detection problem (Steen et al., 2016) and a PCA augmentation technique (Bargoti and Underwood, 2016).

Papers involving simulated data performed additional augmentation techniques such as varying the HSV channels and adding random shadows (Dyrmann et al., 2016b) or adding simulated roots to soil images (Douarre et al., 2016).

4.6. Technical details

From a technical side, almost half of the research works (17 papers, 42%) employed popular CNN architectures such as AlexNet, VGG16 and Inception-ResNet. From the rest, 14 papers developed their own CNN models, 2 papers adopted a first-order Differential Recurrent Neural Networks (DRNN) model, 5 papers preferred to use a Long Short-Term Memory (LSTM) model (Gers et al., 2000), one paper used deep belief networks (DBN) and one paper employed a hybrid of PCA with auto-encoders (AE). Some of the CNN approaches combined their model with a classifier at the output layer, such as logistic regression (Chen et al., 2014), Scalable Vector Machines (SVM) (Douarre et al., 2016), linear regression (Chen et al., 2017), Large Margin Classifiers (LCM) (Xinshao and Cheng, 2015) and macroscopic cellular automata (Song et al., 2016).

Regarding the frameworks used, all the works that employed some well-known CNN architecture had also used a DL framework, with Caffe being the most popular (13 papers, 32%), followed by Tensor Flow (2 papers) and deeplearning4j (1 paper). Ten research works developed their own software, while some authors decided to build their own models on top of Caffe (5 papers), Keras/Theano (5 papers), Keras/TensorFlow (4 papers), Pylearn2 (1 paper), MatConvNet (1 paper) and Deep Learning Matlab Toolbox (1 paper). A possible reason for the wide use of Caffe is that it incorporates various CNN frameworks and datasets, which can be used then easily and automatically by its users.

Most of the studies divided their dataset between training and testing/verification data using a ratio of 80–20 or 90–10 respectively. In addition, various learning rates have been recorded, from 0.001 (Amara et al., 2017) and 0.005 (Mohanty et al., 2016) up to 0.01 (Grinblat et al., 2016). Learning rate is about how quickly a network learns. Higher values help avoid the solver being stuck in local minima, which can reduce performance significantly. A general approach used by many of the evaluated papers is to start out with a high learning rate and lower it as the training goes on. We note that learning rate is very dependent on the network architecture.

Moreover, most of the research works that incorporated popular DL architectures took advantage of transfer learning (Pan and Yang, 2010), which concerns leveraging the already existing knowledge of some related task or domain in order to increase the learning efficiency of the problem under study by fine-tuning pre-trained models. As sometimes it is not possible to train a network from scratch due to having a small training data set or having a complex multi-task network, it is required that the network be at least partially initialized with weights from another pre-trained model. A common transfer learning technique is the use of pre-trained CNN, which are CNN models that have been already trained on some relevant dataset with possibly different number of classes. These models are then adapted to the particular challenge and dataset. This method was followed (among others) in Lu et al. (2017), Douarre et al. (2016), Reyes et al. (2015), Bargoti and Underwood (2016), Steen et al. (2016), Lee et al. (2015), Sa et al. (2016), Mohanty et al. (2016), Christiansen et al. (2016) and Sørensen et al. (2017), for the VGG16, DenseNet, AlexNet and GoogleNet architectures.

4.7. Outputs

Finally, concerning the 31 papers that involved classification, the

classes as used by the authors ranged from 2 (Lu et al., 2017; Pound et al., 2016; Douarre et al., 2016; Milioto et al., 2017) up to 1000 (Reyes et al., 2015). A large number of classes was observed in Luus et al. (2015) (21 land-use classes) (Rebetez et al., 2016) (22 different crops plus soil) (Lee et al., 2015) (44 plant species) and (Xinshao and Cheng, 2015) (91 classes of common weeds found in agricultural fields). In these papers, the number of outputs of the model was equal to the number of classes respectively. Each output was a different probability for the input image, segment, blob or pixel to belong to each class, and then the model picked the highest probability as its predicted class.

From the rest 9 papers, 2 performed predictions of fruits counted (scalar value as output) (Rahneemoonfar and Sheppard, 2017; Chen et al., 2017), 2 identified regions of fruits in the image (multiple bounding boxes) (Bargoti and Underwood, 2016; Sa et al., 2016), 2 predicted animal growth (scalar value) (Demmers et al., 2010, 2012), one predicted weather conditions (scalar value) (Sehgal et al., 2017), one crop yield index (scalar value) (Kuwata and Shibasaki, 2015) and one paper predicted percentage of soil moisture content (scalar value) (Song et al., 2016).

4.8. Performance metrics

Regarding methods used to evaluate performance, various metrics have been employed by the authors, each being specific to the model used at each study. Table 1 lists these metrics, together with their definition/description, and the symbol we use to refer to them in this survey. In some papers where the authors referred to accuracy without specifying its definition, we assumed they referred to classification accuracy (CA, first metric listed in Table 1). From this point onwards, we refer to “DL performance” as its score in some performance metric from the ones listed in Table 1.

CA was the most popular metric used (24 papers, 60%), followed by F1 (10 papers, 25%). Some papers included RMSE (4 papers), IoU (3 papers), RFC (Chen et al., 2017; Rahneemoonfar and Sheppard, 2017) or others. Some works used a combination of metrics to evaluate their efforts. We note that some papers employing CA, F1, P or R, used IoU in order to consider a model's prediction (Bargoti and Underwood, 2016; Sa et al., 2016; Steen et al., 2016; Christiansen et al., 2016; Dyrmann et al., 2017). In these cases, a minimum threshold was put on IoU, and any value above this threshold would be considered as positive by the model.

We note that in some cases, a trade-off can exist between metrics. For example, in a weed detection problem (Milioto et al., 2017), it might be desirable to have a high R to eliminate most weeds, but not eliminating crops is of a critical importance, hence a lower P might be acceptable.

4.9. Overall performance

We note that it is difficult if not impossible to compare between papers, as different metrics are employed for different tasks, considering different models, datasets and parameters. Hence, the reader should consider our comments in this section with some caution.

In 19 out of the 24 papers that involved CA as a metric, accuracy was high (i.e. above 90%), indicating good performance. The highest CA has been observed in the works of Hall et al. (2015), Pound et al. (2016), Chen et al. (2014), Lee et al. (2015), Minh et al. (2017), Potena et al. (2016) and Steen et al., 2016, with values of 98% or more, constituting remarkable results. From the 10 papers using F1 as metric, 5 had values higher than 0.90 with the highest F1 observed in Mohanty et al. (2016) and Minh et al. (2017) with values higher than 0.99.

The works of Dyrmann et al. (2016a), Rußwurm and Körner (2017), Ienco et al. (2017), Mortensen et al. (2016), Rebetez et al. (2016), Christiansen et al. (2016) and Yalcin (2017) were among the ones with the lowest CA (i.e. 73–79%) and/or F1 scores (i.e. 0.558–0.746),

Table 1
Performance metrics used in related work under study.

No.	Performance Metric	Symbol Used	Description
1.	Classification Accuracy	CA	The percentage of correct predictions where the top class (the one having the highest probability), as indicated by the DL model, is the same as the target label as annotated beforehand by the authors. For multi-class classification problems, CA is averaged among all the classes. CA is mentioned as Rank-1 identification rate in (Hall et al., 2015)
2.	Precision	P	The fraction of true positives (TP, correct predictions) from the total amount of relevant results, i.e. the sum of TP and false positives (FP). For multi-class classification problems, P is averaged among the classes. $P = TP / (TP + FP)$
3.	Recall	R	The fraction of TP from the total amount of TP and false negatives (FN). For multi-class classification problems, R gets averaged among all the classes. $R = TP / (TP + FN)$
4.	F1 score	F1	The harmonic mean of precision and recall. For multi-class classification problems, F1 gets averaged among all the classes. It is mentioned as F-measure in (Minh et al., 2017). $F1 = 2 * (TP * FP) / (TP + FP)$
5.	LifeCLEF metric	LC	A score ^a related to the rank of the correct species in the list of retrieved species
6.	Quality Measure	QM	Obtained by multiplying sensitivity (proportion of pixels that were detected correctly) and specificity (which proportion of detected pixels are truly correct). $QM = TP2 / ((TP + FP)(TP + FN))$
7.	Mean Square Error	MSE	Mean of the square of the errors between predicted and observed values
8.	Root Mean Square Error	RMSE	Standard deviation of the differences between predicted values and observed values. A normalized RMSE (N-RMSE) has been used in (Sehgal et al., 2017)
9.	Mean Relative Error	MRE	The mean error between predicted and observed values, in percentage
10.	Ratio of total fruits counted	RFC	Ratio of the predicted count of fruits by the model, with the actual count. The actual count was attained by taking the average count of individuals (i.e. experts or volunteers) observing the images independently
11.	L2 error	L2	Root of the squares of the sums of the differences between predicted counts of fruits by the model and the actual counts
12.	Intersection over Union	IoU	A metric that evaluates predicted bounding boxes, by dividing the area of overlap between the predicted and the ground truth boxes, by the area of their union. An average (Dyrmann et al., 2016b) or frequency weighted (Mortensen et al., 2016) IoU can be calculated
13.	CA-IoU, F1-IoU, P-IoU or R-IoU	CA-IoU F1-IoU P-IoU R-IoU	These are the same CA, F1, P and R metrics as defined above, combined with IoU in order to consider true/false positives/negatives. Used in problems involving bounding boxes. This is done by putting a minimum threshold on IoU, i.e. any value above this threshold would be considered as positive by the metric (and the model involved). Thresholds of 20% (Bargoti and Underwood, 2016), 40% (Sa et al., 2016) and 50% (Steen et al., 2016; Christiansen et al., 2016; Dyrmann et al., 2017) have been observed ^b

^a LifeCLEF 2015 Challenge. <http://www.imageclef.org/lifeclef/2015/plant>.

^b In Appendix B, where we list the values of the metrics used at each paper, we denote CA-IoU(x), F1-IoU(x), P-IoU(x) or R-IoU(x), where x is the threshold (in percentage), over which results are considered as positive by the DL model employed.

however state of the art work in these particular problems has shown lower CA (i.e. SVM, RF, Naïve- Bayes classifier). Particularly in Rußwurm and Körner (2017), the three-unit LSTM model employed provided 16.3% better CA than a CNN, which belongs to the family of DL. Besides, the above can be considered as “harder” problems, because of the use of satellite data (Ienco et al., 2017; Rußwurm and Körner, 2017) large number of classes (Dyrmann et al., 2016a; Rußwurm and Körner, 2017; Rebetez et al., 2016), small training datasets (Mortensen et al., 2016; Christiansen et al., 2016) or very low variation among the classes (Yalcin, 2017; Dyrmann et al., 2016a; Rebetez et al., 2016).

4.10. Generalizations on different datasets

It is important to examine whether the authors had tested their implementations on the same dataset (e.g. by dividing the dataset into training and testing/validation sets) or used different datasets to test their solution. From the 40 papers, only 8 (20%) used different datasets for testing than the one for training. From these, 2 approaches trained their models by using simulated data and tested on real data (Dyrmann et al., 2016b; Rahnemoonfar and Sheppard, 2017) and 2 papers tested their models on a dataset produced 2–4 weeks after, with a more advanced growth stage of plants and weeds (Milioto et al., 2017; Potena et al., 2016). Moreover, 3 papers used different fields for testing than the ones used for training (McCool et al., 2017), with a severe degree of occlusion compared to the other training field (Dyrmann et al., 2017), or containing other obstacles such as people and animals (Steen et al., 2016). Sa et al. (2016) used a different dataset to evaluate whether the model can generalize on different fruits.

From the other 32 papers, different trees were used in training and testing in Chen et al. (2017), while different rooms for pigs (Demmers et al., 2012) and chicken (Demmers et al., 2010) were considered. Moreover, Hall et al. applied condition variations in testing (i.e. translations, scaling, rotations, shading and occlusions) (Hall et al., 2015) while scaling for a certain range translation distance and rotation

angle was performed on the testing dataset in Xinshao and Cheng (2015). The rest 27 papers did not perform any changes between the training/testing datasets, a fact that lowers the overall confidence for the results presented.

Finally, it is interesting to observe how these generalizations affected the performance of the models, at least in cases where both data from same and different datasets were used in testing. In Sa et al. (2016), F1-IoU(40) was higher for the detection of apples (0.938), strawberry (0.948), avocado (0.932) and mango (0.942), than in the default case of sweet pepper (0.838). In Rahnemoonfar and Sheppard (2017), RFC was 2% less in the real images than in the synthetic ones. In Potena et al. (2016), CA was 37.6% less at the dataset involving plants of 4-weeks more advanced growth. According to the authors, the model was trained based on plants that were in their first growth stage, thus without their complete morphological features, which were included in the testing dataset. Moreover, in Milioto et al. (2017) P was 2% higher at the 2-weeks more advanced growth dataset, with 9% lower R.

Hence, in the first case there was improvement in performance (Sa et al., 2016), and in the last three cases a reduction, slight one in Rahnemoonfar and Sheppard (2017) and Milioto et al. (2017) but considerable in Potena et al. (2016). From the other papers using different testing datasets, as mentioned above, high percentages of CA (94–97.3%), P-IoU (86.6%) and low values of MRE (1.8–10%) have been reported. These show that the DL models were able to generalize well to different datasets. However, without more comparisons, this is only a speculation that can be figured out of the small number of observations available.

4.11. Performance comparison with other approaches

A critical aspect of this survey is to examine how DL performs in relation to other existing techniques. The 14th column of Appendix B presents whether the authors of related work compared their DL-based

approach with other techniques used for solving their problem under study. We focus only on comparisons between techniques used for the same dataset at the same research paper, with the same metric.

In almost all cases, the DL models outperform other approaches implemented for comparison purposes. CNN show 1–8% higher CA in comparison to SVM (Chen et al., 2014; Lee et al., 2015; Grinblat et al., 2016; Pound et al., 2016), 41% improvement of CA when compared to ANN (Lee et al., 2015) and 3–8% higher CA when compared to RF (Kussul et al., 2017; Minh et al., 2017; McCool et al., 2017; Potena et al., 2016; Hall et al., 2015). CNN also seem to be superior than unsupervised feature learning with 3–11% higher CA (Luus et al., 2015), 2–44% improved CA in relation to local shape and color features (Dyrmann et al., 2016a; Sørensen et al., 2017), and 2% better CA (Kussul et al., 2017) or 18% less RMSE (Song et al., 2016) compared to multilayer perceptrons. CNN had also superior performance than Penalized Discriminant Analysis (Grinblat et al., 2016), SVM Regression (Kuwata and Shibasaki, 2015), area-based techniques (Rahneemoonfar and Sheppard, 2017), texture-based regression models (Chen et al., 2017), LMC classifiers (Xinshao and Cheng, 2015), Gaussian Mixture Models (Santoni et al., 2015) and Naïve-Bayes classifiers (Yalcin, 2017).

In cases where Recurrent Neural Networks (RNN) (Mandic and Chambers, 2001) architectures were employed, the LSTM model had 1% higher CA than RF and SVM in Ienco et al. (2017), 44% improved CA than SVM in Rußwurm and Körner (2017) and 7–9% better CA than RF and SVM in Minh et al. (2017).

In only one case, DL showed worse performance against another technique, and this was when a CNN was compared to an approach involving local descriptors to represent images together with KNN as the classification strategy (20% worse LC) (Reyes et al., 2015).

5. Discussion

Our analysis has shown that DL offers superior performance in the vast majority of related work. When comparing the performance of DL-based approaches with other techniques at each paper, it is of paramount importance to adhere to the same experimental conditions (i.e. datasets and performance metrics). From the related work under study, 28 out of the 40 papers (70%) performed direct, valid and correct comparisons among the DL-based approach employed and other state-of-art techniques used to solve the particular problem tackled at each paper. Due to the fact that each paper involved different datasets, pre-processing techniques, metrics, models and parameters, it is difficult if not impossible to generalize and perform comparisons between papers. Thus, our comparisons have been strictly limited among the techniques used at each paper. Thus, based on these constraints, we have observed that DL has outperformed traditional approaches used such as SVM, RF, ANN, LMC classifiers and others. It seems that the automatic feature extraction performed by DL models is more effective than the feature extraction process through traditional approaches such as Scale Invariant Feature Transform (SIFT), GLCM, histograms, area-based techniques (ABT), statistics-, texture-, color- and shape-based algorithms, conditional random fields to model color and visual texture features, local de-correlated channel features and other manual feature extraction techniques. This is reinforced by the combined CNN + LSTM model employed in Namin et al. (2017), which outperformed a LSTM model which used hand crafted feature descriptors as inputs by 25% higher CA. Interesting attempts to combine hand-crafted features and CNN-based features were performed in Hall et al. (2015) and Rebetez et al. (2016).

Although DL has been associated with computer vision and image analysis (which is also the general case in this survey), we have observed 5 related works where DL-based models have been trained based on field sensory data (Kuwata and Shibasaki, 2015; Sehgal et al., 2017) and a combination of static and dynamic environmental variables (Song et al., 2016; Demmers et al., 2010, 2012). These papers indicate the potential of DL to be applied in a wide variety of agricultural problems,

not only those involving images.

Examining agricultural areas where DL techniques have been applied, leaf classification, leaf and plant disease detection, plant recognition and fruit counting have some papers which present very good performance (i.e. CA > 95%, F1 > 0.92 or RFC > 0.9). This is probably because of the availability of datasets in these domains, as well as the distinct characteristics of (sick) leaves/plants and fruits in the image. On the other hand, some papers in land cover classification, crop type classification, plant phenology recognition and weed detection showed average performance (i.e. CA < 87% or F1 < 0.8). This could be due to leaf occlusion in weed detection, use of noise-prone satellite imagery in land cover problems, crops with low variation and botanical relationship or the fact that appearances change very gradually while plants grow in phenology recognition efforts.

Without underestimating the quality of any of the surveyed papers, we highlight some that claim high performance (CA > 91%, F1-IoU (20) > 0.90 or RFC > 0.91), considering the complexity of the problem in terms of its definition or the large number of classes involved (more than 21 classes). These papers are the following: (Mohanty et al., 2016; Luus et al., 2015; Lee et al., 2015; Rahneemoonfar and Sheppard, 2017; Chen et al., 2017; Bargoti and Underwood, 2016; Xinshao and Cheng, 2015 and Hall et al., 2015). We also highlight papers that trained their models on simulated data, and tested them on real data, which are (Dyrmann et al., 2016b; Rahneemoonfar and Sheppard, 2017 and Douarre et al., 2016). These works constitute important efforts in the DL community, as they attempt to solve the problem of inexistent or not large enough datasets in various problems.

Finally, as discussed in Section 4.10, most authors used the same datasets for training and testing their implementation, a fact that lowers the confidence in the overall findings, although there have been indications that the models seem to generalize well, with only small reductions in performance.

5.1. Advanced deep learning applications

Although the majority of papers used typical CNN architectures to perform classification (23 papers, 57%), some authors experimented with more advanced models in order to solve more complex problems, such as crop type classification from UAV imagery (CNN + HistNN using RGB histograms) (Rebetez et al., 2016), estimating number of tomato fruits (Modified Inception-ResNet CNN) (Rahneemoonfar and Sheppard, 2017) and estimating number of orange or apple fruits (CNN adapted for blob detection and counting + Linear Regression) (Chen et al., 2017). Particularly interesting were the approaches employing the Faster Region-based CNN + VGG16 model (Bargoti and Underwood, 2016; Sa et al., 2016), in order not only to count fruits and vegetables, but also to locate their placement in the image by means of bounding boxes. Similarly, the work in (Dyrmann et al., 2017) used the DetectNet CNN to detect bounding boxes of weed instances in images of cereal fields. These approaches (Faster Region-based CNN, DetectNet CNN) constitute a very promising research direction, since the task of identifying the bounding box of fruits/vegetables/weeds in an image has numerous real-life applications and could solve various agricultural problems.

Moreover, considering not only space but also time series, some authors employed RNN-based models in land cover classification (one-unit LSTM model + SVM) (Ienco et al., 2017), crop type classification (three-unit LSTM) (Rußwurm and Körner, 2017), classification of different accessions of *Arabidopsis thaliana* based on successive top-view images (CNN + LSTM) (Namin et al., 2017), mapping winter vegetation quality coverage (Five-unit LSTM, Gated Recurrent Unit) (Minh et al., 2017), estimating the weight of pigs or chickens (DRNN) (Demmers et al., 2010, 2012) and for predicting weather based on previous year's conditions (LSTM) (Sehgal et al., 2017). RNN-based models offer higher performance, as they can capture the time dimension, which is impossible to be exploited by simple CNN. RNN architectures tend to

Table 2
Difference in performance between CNN and RNN.

No.	Application in Agriculture	Performance Metric	Difference in Performance	Reference
1.	Crop type classification considering time series	CA, F1	Three-unit LSTM: 76.2% (CA), 0.558 (F1) CNN: 59.9% (CA), 0.236 (F1)	Rußwurm and Körner (2017)
2.	Classify the phenotyping of Arabidopsis in four accessions	CA	CNN + LSTM: 93% CNN: 76.8%	Namin et al. (2017)

exhibit dynamic temporal behavior, being able to record long-short temporal dependencies, remembering and forgetting after some time or when needed (i.e. LSTM). Differences in performance between RNN and CNN are distinct in the related work under study, as shown in Table 2. This 16% improvement in CA could be attributed to the additional information provided by the time series. For example, in the crop type classification case (Rußwurm and Körner, 2017), the authors mention, “crops change their spectral characteristics due to environmental influences and can thus not be monitored effectively with classical mono-temporal approaches. Performance of temporal models increases at the beginning of vegetation period”. LSTM-based approaches work well also for low represented and difficult classes, as demonstrated in Ienco et al. (2017).

Finally, the critical aspect of fast processing of DL models in order to be easily used in robots for real-time decision making (e.g. detection of weeds) was examined in McCool et al. (2017), and it is worth-mentioning. The authors have showed that a lightweight implementation had only a small penalty in CA (3.90%), being much faster (i.e. processing of 40.6 times more pixels per second). They proposed the idea of “teacher and student networks”, where the teacher is the more heavy approach that helps the student (light implementation) to learn faster and better.

5.2. Advantages of deep learning

Except from improvements in performance of the classification/prediction problems in the surveyed works (see Sections 4.9 and 4.11), the advantage of DL in terms of reduced effort in feature engineering was demonstrated in many of the papers. Hand-engineered components require considerable time, an effort that takes place automatically in DL. Besides, sometimes manual search for good feature extractors is not an easy and obvious task. For example, in the case of estimating crop yield (Kuwata and Shibasaki, 2015), extracting manually features that significantly affected crop growth was not possible. This was also the case of estimating the soil moisture content (Song et al., 2016).

Moreover, DL models seem to generalize well. For example, in the case of fruit counting, the model learned explicitly to count (Rahneemoonfar and Sheppard, 2017). In the banana leaf classification problem (Amara et al., 2017), the model was robust under challenging conditions such as illumination, complex background, different resolution, size and orientation of the images. Also in the fruits counting papers (Chen et al., 2017; Rahneemoonfar and Sheppard, 2017), the models were robust to occlusion, variation, illumination and scale. The same detection frameworks could be used for a variety of circular fruits such as peaches, citrus, mangoes etc. As another example, a key feature of the DeepAnomaly model was the ability to detect unknown objects/anomalies and not just a set of predefined objects, exploiting the homogeneous characteristics of an agricultural field to detect distant, heavy occluded and unknown objects (Christiansen et al., 2016). Moreover, in the 8 papers mentioned in Section 4.10 where different datasets were used for testing, the performance of the model was generally high, with only small reductions in performance in comparison with the performance when using the same dataset for training and testing.

Although DL takes longer time to train than other traditional

approaches (e.g. SVM, RF), its testing time efficiency is quite fast. For example, in detecting obstacles and anomaly (Christiansen et al., 2016), the model took much longer to train, but after it did, its testing time was less than the one of SVM and KNN. Besides, if we take into account the time needed to manually design filters and extract features, “the time used on annotating images and training the CNN becomes almost negligible” (Sørensen et al., 2017).

Another advantage of DL is the possibility to develop simulated datasets to train the model, which could be properly designed in order to solve real-world problems. For example, in the issue of detecting weeds and maize in fields (Dyrmann et al., 2016b), the authors overcame the plant foliage overlapping problem by simulating top-down images of overlapping plants on soil background. The trained network was then capable of distinguish weeds from maize even in overlapping conditions.

5.3. Disadvantages and limitations of deep learning

A considerable drawback and barrier in the use of DL is the need of large datasets, which would serve as the input during the training procedure. In spite of data augmentation techniques which augment some dataset with label-preserving transformations, in reality at least some hundreds of images are required, depending on the complexity of the problem under study (i.e. number of classes, precision required etc.). For example, the authors in Mohanty et al. (2016) and Sa et al. (2016) commented that a more diverse set of training data was needed to improve CA. A big problem with many datasets is the low variation among the different classes (Yalcin, 2017), as discussed in Section 4.3, or the existence of noise, in the form of low resolution, inaccuracy of sensory equipment (Song et al., 2016), crops’ occlusions, plants overlapping and clustering, and others.

As data annotation is a necessary operation in the large majority of cases, some tasks are more complex and there is a need for experts (who might be difficult to involve) in order to annotate input images. As mentioned in Amara et al. (2017), there is a limited availability of resources and expertise on banana pathology worldwide. In some cases, experts or volunteers are susceptible to errors during data labeling, especially when this is a challenging task e.g. fruit count (Chen et al., 2017; Bargoti and Underwood, 2016) or to determine if images contain weeds or not (Sørensen et al., 2017; Dyrmann et al., 2017).

Another limitation is the fact that the DL models can learn some problem particularly well, even generalize in some aspects as mentioned in Section 5.2, but they cannot generalize beyond the “boundaries of the dataset’s expressiveness”. For example, classification of single leaves, facing up, on a homogeneous background is performed in Mohanty et al. (2016). A real world application should be able to classify images of a disease as it presents itself directly on the plant. Many diseases do not present themselves on the upper side of leaves only. As another example, plant recognition in Lee et al. (2015) was noticeably affected by environmental factors such as wrinkled surface and insect damages. The model for counting tomatoes in Rahneemoonfar and Sheppard (2017) could count ripe and half-ripe fruits, however, “it failed to count green fruits because it was not trained for this purpose”. If an object size in a testing image was significantly less than that of a

training set, the model missed the detection in [Sa et al. \(2016\)](#). Difficulty in detecting heavily occluded and distant objects was observed in [Christiansen et al. \(2016\)](#). Occlusion was a serious issue also in [Hall et al. \(2015\)](#).

A general issue in computer vision, not only in DL, is that data pre-processing is sometimes a necessary and time-consuming task, especially when satellite or aerial photos are involved, as we saw in Section 4.4. A problem with hyperspectral data is their high dimensionality and limited training samples ([Chen et al., 2014](#)).

Moreover, sometimes the existing datasets do not describe completely the problem they target ([Song et al., 2016](#)). As an example, for estimating corn yield ([Kuwata and Shibasaki, 2015](#)), it was necessary to consider also external factors other than the weather by inputting cultivation information such as fertilization and irrigation.

Finally, in the domain of agriculture, there do not exist many publicly available datasets for researchers to work with, and in many cases, researchers need to develop their own sets of images. This could require many hours or days of work.

5.4. Future of deep learning in agriculture

Observing [Appendix A](#), which lists various existing applications of computer vision in agriculture, we can see that only the problems of land cover classification, crop type estimation, crop phenology, weed detection and fruit grading have been approximated using DL. It is interesting to see how DL would behave also in other agricultural-related problems listed in [Appendix A](#), such as seeds identification, soil and leaf nitrogen content, irrigation, plants' water stress detection, water erosion assessment, pest detection, herbicide use, identification of contaminants, diseases or defects on food, crop hail damage and greenhouse monitoring. Intuitively, since many of the aforementioned research areas employ data analysis techniques (see [Appendix A](#)) with similar concepts and comparable performance to DL (i.e. linear and logistic regression, SVM, KNN, K-means clustering, Wavelet-based filtering, Fourier transform) ([Singh et al., 2016](#)), then it could be worth to examine the applicability of DL on these problems too.

Other possible application areas could be the use of aerial imagery (i.e. by means of drones) to monitor the effectiveness of the seeding process, to increase the quality of wine production by harvesting grapes at the right moment for best maturity levels, to monitor animals and their movements to consider their overall welfare and identify possible diseases, and many other scenarios where computer vision is involved.

In spite of the limited availability of open datasets in agriculture, in [Appendix C](#), we list some of the most popular, free to download datasets available on the web, which could be used by researchers to start testing their DL architectures. These datasets could be used to pre-train DL models and then adapt them to more specific future agricultural challenges. In addition to these datasets, remote sensing data containing multi-temporal, multi-spectral and multi-source images that could be used in problems related to land and crop cover classification are available from satellites such as MERIS, MODIS, AVHRR, RapidEye, Sentinel, Landsat etc.

More approaches adopting LSTM or other RNN models are expected in the future, exploiting the time dimension to perform higher performance classification or prediction. An example application could be to estimate the growth of plants, trees or even animals based on previous consecutive observations, to predict their yield, assess their water needs or avoid diseases from occurring. These models could find applicability in environmental informatics too, for understanding climatic change, predicting weather conditions and phenomena, estimating the environmental impact of various physical or artificial processes ([Kamilaris et al., 2017a](#)) etc. Related work under study involved up to a five-unit LSTM model ([Minh et al., 2017](#)). We expect in the future to see more layers stacked together in order to build more complex LSTM

architectures ([Ienco et al., 2017](#)). We also believe that datasets with increasing temporal sequence length will appear, which could improve the performance of LSTM ([Rußwurm and Körner, 2017](#)).

Moreover, more complex architectures would appear, combining various DL models and classifiers together, or combining hand-crafted features with automatic features extracted by using various techniques, fused together to improve the overall outcome, similar to what performed in [Hall et al. \(2015\)](#) and [Rebetez et al. \(2016\)](#). Researchers are expected to test their models using more general and realistic dataset, demonstrating the ability of the models to generalize to various real-world situations. A combination of popular performance metrics, such as the ones mentioned in [Table 1](#), are essential to be adopted by the authors for comparison purposes. It would be desirable if researchers made their datasets publicly available, for use also by the general research community.

Finally, some of the solutions discussed in the surveyed papers could have a commercial use in the near future. The approaches incorporating Faster Region-based CNN and DetectNet CNN ([Bargoti and Underwood, 2016](#); [Chen et al., 2017](#); [Rahnemounfar and Sheppard, 2017](#)) would be extremely useful for automatic robots that collect crops, remove weeds or for estimating the expected yields of various crops. A future application of this technique could be also in microbiology for human or animal cell counting ([Chen et al., 2017](#)). The DRNN model controlling the daily feed intake of pigs or chicken, predicting quite accurately the required feed intake for the whole of the growing period, would be useful to farmers when deciding on a growth curve suitable for various scenarios. Following some growth patterns would have potential advantages for animal welfare in terms of leg health, without compromising the idea animals' final weight and total feed intake requirement ([Demmers et al., 2010, 2012](#)).

6. Conclusion

In this paper, we have performed a survey of deep learning-based research efforts applied in the agricultural domain. We have identified 40 relevant papers, examining the particular area and problem they focus on, technical details of the models employed, sources of data used, pre-processing tasks and data augmentation techniques adopted, and overall performance according to the performance metrics employed by each paper. We have then compared deep learning with other existing techniques, in terms of performance. Our findings indicate that deep learning offers better performance and outperforms other popular image processing techniques. For future work, we plan to apply the general concepts and best practices of deep learning, as described through this survey, to other areas of agriculture where this modern technique has not yet been adequately used. Some of these areas have been identified in the discussion section.

Our aim is that this survey would motivate more researchers to experiment with deep learning, applying it for solving various agricultural problems involving classification or prediction, related to computer vision and image analysis, or more generally to data analysis. The overall benefits of deep learning are encouraging for its further use towards smarter, more sustainable farming and more secure food production.

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Appendix A. Applications of computer vision in agriculture and popular techniques used

No.	Application in Agriculture	Remote sensing	Techniques for data analysis
1.	Soil and vegetation/crop mapping	Hyperspectral imaging (satellite and airborne), multi-spectral imaging (satellite), synthetic aperture radar (SAR)	Image fusion, SVM, end-member extraction algorithm, co-polarized phase differences (PPD), linear polarizations (HH, VV, HV), distance-based classification, decision trees, linear mixing models, logistic regression, ANN, NDVI
2.	Leaf area index and crop canopy	Hyperspectral imaging (airborne), multi-spectral imaging (airborne)	Linear regression analysis, NDVI
3.	Crop phenology	Satellite remote sensing (general)	Wavelet-based filtering, Fourier transforms, NDVI
4.	Crop height, estimation of yields, fertilizers' effect and biomass	Light Detection and Ranging (LIDAR), hyperspectral and multi-spectral imaging, SAR, red-edge camera, thermal infrared	Linear and exponential regression analysis, linear polarizations (VV), wavelet-based filtering, vegetation indices (NDVI, ICWSI), ANN
5.	Crop monitoring	Satellite remote sensing (hyperspectral and multi-spectral imaging), NIR camera, SAR	Stepwise discriminant analysis (DISCRIM) feature extraction, linear regression analysis, co-polarized phase differences (PPD), linear polarizations (HH, VV, HV, RR and RL), classification and regression tree analysis
6.	Identification of seeds and reorganization of species	Remote sensing in general, cameras and photo-detectors, hyperspectral imaging	Principal component analysis, feature extraction, linear regression analysis
7.	Soil and leaf nitrogen content and treatment, salinity detection	Hyperspectral and multi-spectral imaging, thermal imaging	Linear and exponential regression analysis
8.	Irrigation	Satellite remote sensing (hyperspectral and multi-spectral imaging), red-edge camera, thermal infrared	Image classification techniques (unsupervised clustering, density slicing with thresholds), decision trees, linear regression analysis, NDVI
9.	Plants water stress detection and drought conditions	Satellite remote sensing (hyperspectral and multi-spectral imaging, radar images), thermal imaging, NIR camera, red-edge camera	Fraunhofer Line Depth (FLD) principle, linear regression analysis, NDVI
10.	Water erosion assessment	Satellite remote sensing (optical and radar images), SAR, NIR camera	Interferometric SAR image processing, linear and exponential regression analysis, contour tracing, linear polarizations (HH, VV)
11.	Pest detection and management	Hyperspectral and multi-spectral imaging, microwave remote sensing, thermal camera	Image processing using sample imagery, linear and exponential regression analysis, statistical analysis, CEM nonlinear signal processing, NDVI
12.	Weed detection	Remote sensing in general, optical cameras and photo-detectors, hyperspectral and multi-spectral imaging	Pixel classification based on k-means clustering and Bayes classifier, feature extraction techniques with FFT and GLCM, wavelet-based classification and Gabor filtering, genetic algorithms, fuzzy techniques, artificial neural networks, erosion and dilation segmentation, logistic regression, edge detection, color detection, principal component analysis
13.	Herbicide	Remote sensing in general, optical cameras and photo-detectors	Fuzzy techniques, discriminant analysis
14.	Fruit grading	Optical cameras and photo-detectors, monochrome images with different illuminations	K-means clustering, image fusion, color histogram techniques, machine learning (esp. SVM), Bayesian discriminant analysis, Bayes filtering, linear discriminant analysis
15.	Packaged food and food products – identification of contaminants, diseases or defects, bruise detection	X-ray imaging (or transmitted light), CCD cameras, monochrome images with different illuminations, thermal cameras, multi-spectral and hyperspectral NIR-based imaging	3D vision, invariance, pattern recognition and image modality, multivariate image analysis with principal component analysis, K-mean clustering, SVM, linear discriminant analysis, classification trees, K-nearest neighbors, decision trees, fusion, feature extraction techniques with FFT, standard Bayesian discriminant analysis, feature analysis, color, shape and geometric features using discrimination analysis, pulsed-phase thermography
16.	Crop hail damage	Multi-spectral imaging, polarimetric radar imagery	Linear and exponential regression analysis, unsupervised image classification
17.	Agricultural expansion and intensification	Satellite remote sensing in general	Wavelet-based filtering
18.	Greenhouse monitoring	Optical and thermal cameras	Linear and exponential regression analysis, unsupervised classification, NDVI, IR thermography

Appendix B. Applications of deep learning in agriculture

No.	Agri Area	Problem Description	Data Used	Classes and Labels	Variation among Classes	DL Model Used
1.	Leaf classification	Classify leaves of different plant species	Flavia dataset, consisting of 1907 leaf images of 32 species with at least 50 images per species and at most 77 images.	32 classes: 32 Different plant species	N/A	Author-defined CNN + RF classifier
2.	Leaf disease detection	13 different types of plant diseases out of healthy leaves	Authors-created database containing 4483 images.	15 classes: Plant diseases (13), healthy leaves (1) and background images (1)	N/A	CaffeNet CNN
3.	Plant disease detection	Identify 14 crop species and 26 diseases	PlantVillage public dataset of 54,306 images of diseased and healthy plant leaves collected under controlled conditions.	38 class labels as crop-disease pairs	N/A	AlexNet, GoogleNet CNNs
4.		Classify banana leaves' diseases	Dataset of 3700 images of banana diseases obtained from the PlantVillage dataset.	3 classes: healthy, black sigatoka and black speckle	N/A	LeNet CNN
5.	Land cover classification	Identify 13 different land-cover classes in KSC and 9 different classes in Pavia	A mixed vegetation site over Kennedy Space Center (KSC), FL, USA (Dataset 1), and an urban site over the city of Pavia, Italy (Dataset 2). Hyperspectral datasets.	13 different land-cover classes (Dataset 1), 9 land cover classes trees (Dataset 2): Soil, meadow, water, shadows, different materials	N/A	Hybrid of PCA, autoencoder (AE), and logistic regression
6.		Identify 21 land-use classes containing a variety of spatial patterns	UC Merced land-use data set. Aerial orthoimagery with a 0.3048-m pixel resolution. Dataset compiled from a selection of 100 images/class.	21 land-use classes: Agricultural, airplane, sports, beach, buildings, residential, forest, freeway, harbor, parking lot, river etc.	High relevance between medium density and dense residential, as well as between buildings and storage tanks	Author-defined CNN + multiview model averaging
7.		Extract information about cultivated land	Images from UAV at the areas Pengzhou County and Guanghan County, Sichuan Province, China.	2 classes: Cultivated vs. non-cultivated	The cultivated land samples and part of forest land samples were easily confused	Author-defined CNN
8.		Land cover classification considering time series	First dataset generated using a time series of Pléiades VHRS images at THAU Basin. Second dataset generated from an annual time series of 23 Landsat 8 images acquired in 2014 above Reunion Island.	11 classes (dataset 1), 9 classes (dataset 2). Land cover classes such as trees, crops, forests, water, soils, urban areas, grasslands, etc. (Image object or pixel)	Tree Crops, Summer crops and Truck Farming were classes highly mixed	One-unit LSTM + RFF, One-unit LSTM + SVM
9.	Crop type classification	Classification of crops wheat, maize, soybeans sunflower and sugar beet	19 multi-temporal scenes acquired by Landsat-8 and Sentinel-1A RS satellites from a test site in Ukraine.	11 classes: water, forest, grassland, bare land, wheat, maize, rapeseed, cereals, sugar beet, sunflowers and soybeans.	General confusion between maize and soybeans	Author-defined CNN
10.		Classification of crops oil radish, barley, seeded grass, weed and stump	36 plots at Foulum Research Center, Denmark containing oil radish as a catch crop and amounts of barley, grass, weed and stump. 352 patches in total.	7 classes: oil radish, barley, weed, stump, soil, equipment and unknown (pixel of the image)	Coarse features (radish leafs and soil) were predicted quite well. Finer features (barley, grass or stump) not so much.	Adapted version of VGG16 CNN
11.		Crop type classification considering time series	A raster dataset of 26 SENTINEL 2A images, acquired between 2015	19 classes: corn, meadow, asparagus, rape, hop, summer oats, winter spelt, fallow, wheat, barley, winter rye, beans and others	Some classes represent distinct cultivated crops, others (such as meadow, fallow, triticale, wheat, and rye) are botanically related.	Three-unit LSTM
			2016 at Munich Germany. Shortwave infrared 1 and 2 bands were selected.			

12.	Crop type classification from UAV imagery	Aerial images of experimental farm fields issued from a series of experiments conducted by the Swiss Confederation's Agroscope research center.	23 classes: 22 different crops plus soil (pixel of the image)	Lin and Simplex have very similar histograms	CNN + HistNN (using RGB histograms)
13.	Plant recognition	Recognize 7 views of different plants: entire plant, branch, flower, fruit, leaf, stem and scans	LifeCLEF 2015 plant dataset, which has 91,759 images distributed in 13,887 plant observations. Each observation captures the appearance of the plant from various points of view: entire plant, leaf branch, fruit, stem scan, flower.	1000 classes: Species that include trees, herbs, and ferns, among others.	AlexNet CNN
14.		Root and shoot feature identification and localisation	The first dataset contains 2500 annotated images of whole root systems. The second hand-annotated 1664 images of wheat plants, labeling leaf tips, leaf bases, ear tips, and ear bases.	2 classes: Prediction if a root tip is present or not (first dataset) 5 classes: Leaf tips and bases, ear tips and bases, and negative (second dataset)	Author-defined CNN
15.		Recognize 44 different plant species	MalayaKew (MK) Leaf Dataset which consists of 44 classes, collected at the Royal Botanic Gardens, Kew, England.	44 classes: Species such as acutissima, macranthera, rubra, robur f. purpurascens etc.	AlexNet CNN
16.		Identify plants from leaf vein patterns of white, soya and red beans	866 leaf images provided by INTA Argentina. Dataset divided into three classes: 422 images correspond to soybean leaves, 272 to red bean leaves and 172 to white bean leaves.	3 classes: Legume species white bean, red bean and soybean	Author-defined CNN
17.	Plant phenology recognition	Classify phenological stages of several types of plants purely based on the visual data	Dataset collected through TARBIL Agro-informatics Research Center of ITU, for which over a thousand agrostations are placed throughout Turkey. Different images of various plants, at different phenological stages.	9 classes: Different growth stages of plants, starting from plowing to cropping, for the plants wheat, barley, lentil, cotton, pepper and corn. (image segment)	AlexNet CNN
18.		Classify the phenotyping of Arabidopsis in four accessions	Dataset composed of sequences of images captured from the plants in different days while they grow, successive top-view images of different accessions of Arabidopsis thaliana.	4 classes: 4 different accessions of Arabidopsis: Genotype states SF-2, CVI, Landsberg (Ler) and Columbia (Col)	CNN + LSTM
19.	Segmentation of root and soil	Identify roots from soils	Soil images coming from X-ray tomography.	2 classes: Root or soil (pixel of the image)	Author-defined CNN with SVM for classification
20.	Crop yield estimation	Estimate corn yield of county level in U.S.	Corn yields from 2001 to 2010 in Illinois U.S., downloaded from Climate Research Unit (CRU), plus MODIS Enhanced Vegetation Index.	Crop yield index (scalar value)	Author-defined CNN
21.		Mapping winter vegetation quality coverage considering time series	Sentinel-1 dataset including 13 acquisitions in TOPS mode from October 2016 to February 2017, with a temporal baseline of 12 days. Dual-polarization (VV + VH) data in 26 images.	5 classes: Estimations of the quality of vegetative development as bare soil, very low, low, average, high	Five-unit LSTM, Gated Recurrent Unit (GRU)
22.	Fruit counting	Predict number of tomatoes in the images	24,000 synthetic images produced by the authors.	Estimated number of tomato fruits (scalar value)	Modified Inception-ResNet CNN

23.	Map from input images of apples and oranges to total fruit counts	71 1280 × 960 orange images (day time) and 21 1920 × 1200 apple images (night time).	Number of orange or apple fruits (scalar value)	High variation in CA. For orange, dataset has high occlusion, depth variation, and uncontrolled illumination. For apples, data set has high color similarity between fruit/ foliage	CNN (blob detection and counting) + Linear Regression
24.	Fruit detection in orchards, including mangoes, almonds and apples	Images of three fruit varieties: apples (726), almonds (385) and mangoes (1154), captured at orchards in Victoria and Queensland, Australia.	Sections of apples, almonds and mangoes at the image (bounding box)	Within class variations due to distance to fruit illumination, fruit clustering, and camera view-point. Almonds similar in color and texture to the foliage	Faster Region-based CNN with VGG16 model
25.	Detection of sweet pepper and rock melon fruits	122 images obtained from two modalities: color (RGB) and Near-Infrared (NIR).	Sections of sweet red peppers and rock melons on the image (bounding box)	Variations to camera setup, time and locations of data acquisition. Time for data collection is day and night, sites are different. Varied fruit ripeness.	Faster Region-based CNN with VGG16 model
26.	Obstacle detection	Identify ISO barrel-shaped obstacles in row crops and grass mowing	Identify if a barrel-shaped object is present in the image (bounding box)	N/A	AlexNet CNN
27.	Detect obstacles that are distant, heavily occluded and unknown	Background data of 48 images and test data of 48 images from annotations of humans, houses, barrels, wells and mannequins.	Classify each pixel as either foreground (contains a human) or background (anomaly detection)	N/A	AlexNet and VGG CNNs
28.	Identification of weeds	Classify 91 weed seed types	Dataset of 3980 images containing 91 types of weed seeds.	91 classes: Different common weeds found in agricultural fields	PCANet + LMC classifiers
29.	Classify weed from crop species based on 22 different species in total.	Dataset of 10,413 images, taken mainly from BBCH 12–16 containing 22 weed and crop species at early growth stages.	22 classes: Different species of weeds and crops at early growth stages e.g. chamomile, knotweed, cranesbill, chickweed and veronica	Similarity between some classes is very high (only slight differences in shape, texture, and color)	Variation of VGG16
30.	Identify thistle in winter wheat and spring barley images	4500 images from 10, 20, 30, and 50 m of altitude captured by a Canon PowerShot G15 camera.	2 classes: Whether the image contains thistle in winter wheat or not (Heatmap of classes is generated at the output)	Small variations in some images depending on the percentage of thistles they contain	DenseNet CNN
31.	Weed segmentation for robotic platforms	Crop/Weed Field Image Dataset (CW-FID), consists of 20 training and 40 testing images. A dataset of 60 top-down field images of a common culture (organic carrots) with the presence of intra-row and close-to-crop weeds.	2 classes: carrot plants and weeds (image region)	N/A	Adapted version of Inception-v3 + lightweight DCNN + set of K lightweight models as a mixture model (MixDCNN)
32.	Automating weed detection in color images despite heavy leaf occlusion	1427 images from winter wheat fields, of which 18,541 weeds have been annotated, collected using a camera mounted on an all-terrain vehicle.	Detect single weed instances in images of cereal fields (bounding box). A coverage map is produced.	Large parts of the weeds overlap with wheat plants	Based on DetectNet CNN (which is based on GoogLeNet CNN)
33.	Crop/weed detection and classification	1969 RGB + NIR images captured using a JAI camera in nadir view placed on a UAV.	Identify if an image patch belongs to weed or sugar beet (image region)	N/A	Author-defined CNN
34.				N/A	Author-defined CNN

	Detecting and classifying sugar beet plants and weeds	1600 4-channels RGB + NIR images captured before (700 images) and after (900 images) a 4-week period, provided by a multispectral JAI camera mounted on a BOSCH Bonirob farm robot.	Identifies if a blob belongs to sugar beet crop, weeds or soil (image blob)						Adapted version of VGG16 CNN
35.	Detecting and classifying weeds and maize in fields	Simulated top-down images of overlapping plants on soil background A total of 301 images of soil and 8430 images of segmented plants. The plants cover 23 different weed species and maize.	Identifies if an image patch belongs to weed, soil or maize crop (image pixel)	N/A					
36.	Prediction of soil moisture content	Soil data collected from an irrigated corn field (an area of 22 sq. km) in the Zhangye oasis, Northwest China.	Percentage of soil moisture content (SMC) (scalar value)	N/A					Deep belief network-based macroscopic cellular automata (DBN-MCA) GLCM – CNN
37.	Animal research	1300 images collected by the authors.	5 classes: Cattle races, Bali Onggole or Pasuruan, Aceh Madura and Pesisir	N/A					
38.	Predict growth of pigs	160 pigs, housed in two climate controlled rooms, four pens/room, 10 pigs/pen. Ammonia, ambient and indoor air temperature and humidity, feed dosage and ventilation measured at 6-minute intervals.	Estimation of the weight of pigs (scalar value)	N/A					First-order DRNN
39.	Control of the growth of broiler chickens	Collecting data from 8 rooms, each room housing 262 broilers, measuring bird weight, feed amount, light intensity and relative humidity.	Estimation of the weight of chicken (scalar value)	N/A					First-order DRNN
40.	Weather prediction	Syngenta Crop Challenge 2016 dataset, containing 6490 sub-regions with three weather condition attributes from the years 2000 to 2015.	Predicted values of temperature, precipitation and solar radiation (scalar value)	N/A					LSTM
No.	FW Used	Data Pre-Processing	Data	Data for Training vs. Testing augmentation	Performance Metric Used	Value of Metric Used	Comparison with other technique	Ref.	
1.	Caffe	Feature extraction based on Histograms of Curvature over Scale (HoCS), shape and statistical features, use of normalized excessive green (NExG) vegetative index, white border doubling image size, segmentation	N/A	Same. (condition variations applied in testing: translations, scaling, rotations, shading and occlusions)	CA	97.3% ± 0.6%	Feature extraction (shape and statistical features) and RF classifier (91.2% ± 1.6%)	Hall et al. (2015)	
2.	Caffe	Cropping, square around the leaves to highlight region of	Affine transform (translation, rotation),	Same	CA	96.30%	Better results than SVM (no more details)	Sladojevic et al. (2016)	

3.	Caffe	interest, resized to 256×256 pix, dupl. image removal Resized to 256×256 pix., segmentation, background Information removal, fixed color casts	perspective transform, and image rotations. N/A	Same. Also tested on a dataset of downloaded images from Bing Image Search and IPM Images	F1	0.9935	Substantial margin in standard benchmarks with approaches using hand-engineered features	Mohanty et al. (2016)
4.	deeplearning4j	Resized to 60×60 pix., converted to grayscale	N/A	Same	CA, F1	96+ % (CA), 0.968 (F1)	Methods using hand-crafted features not generalize well	Amara et al. (2017)
5.	Developed by the authors	Some bands removed due to noise	N/A	Same	CA	98.70%	1% more precise than RBF-SVM	Chen et al. (2014)
6.	Theano	From RGB to HSV (hue-saturation-value) color model, resized to 96×96 pix., creation of multiscale views	Views flipped horizontally or vertically with a probability of 0.5	Same	CA	93.48%	Unsupervised feature learning (UFL): 82–90% SIFT : 85%	Luus et al. (2015)
7.	N/A	Orthorectification, image matching, linear land elimination, correct distortion, zoomed to 40×40 pix.	N/A	Same	CA	88–91%	N/A	Lu et al. (2017)
8.	Keras/Theano	Multiresolution segmentation technique, feature extraction, pixel-wise multi-temporal linear interpolation, various radiometric indices calculated	N/A	Same	CA, F1	First Dataset: 75.34% (CA), 0.7463 (F1) Second Dataset: 84.61% (CA), 0.8441 (F1)	RF and SVM (best of both): First Dataset: 74.20% (CA), 0.7158 (F1) Second Dataset: 83.82% (CA), 0.8274 (F1)	Ienco et al. (2017)
9.	Developed by the authors	Calibration, multi-looking, speckle filtering (3×3 window with Refined Lee algorithm), terrain correction, segmentation, restoration of missing data	N/A	Same	CA	94.60%	Multilayer perceptron : 92.7%, RF : 88%	Kussul et al. (2017)
10.	Developed by the authors	Resized to 1600×1600 pix. centered on the sample areas, division into 400×400 pix. patches	Rotations 0, 90, 180 and 270 degrees, flipped diagonally and same set of rotations	Same	CA, IoU	79% (CA), 0.66 (IoU)	N/A	Mortensen et al. (2016)

11. TensorFlow	Atmospherically corrected	N/A	Same	CA, F1	76.2% (CA), 0.558 (F1)	CNN: 59.9% (CA), 0.236 (F1)	Rußwurm and Körner (2017)
12. Keras	Image segmentation	N/A	Same	F1	0.90 (experiment 0), 0.73 (experiment 1)	SVM: 31.7 (CA), 84.8% 0.317 (F1) CNN: 0.83 (experiment 0), 0.70 (experiment 1) HistNN: 0.86 (experiment 0), 0.71 (experiment 1)	Rebetez et al. (2016)
13. Caffe	N/A	N/A	Same	LC	48.60%	20% worse than local descriptors to represent images and KNN, dense SIFT and a Gaussian Mixture Model	Reyes et al. (2015)
14. Caffe	Image cropping at annotated locations 128×128 pix., resized to 64×64 for use in the network	N/A	Same	CA	98.4% (first dataset) 97.3% (second dataset)	Sparse coding approach using SIFT + SVM: 80–90%	Pound et al. (2016)
15. Caffe	Foreground pixels extracted using HSV color space, image cropping within leaf area	Rotation in 7 different orientations	Same	CA	99.60%	SVM: 95.1%, ANN: 58%	Lee et al. (2015)
16. Pylearn2	Vein segmentation, central patch extraction	N/A	Same	CA	96.90%	Penalized Discriminant Analysis (PDA): 95.1% SVM and RF slightly worse	Grinblat et al. (2016)
17. Developed by the authors	Image segmentation	Images are divided into large patches and features are extracted for each patch. 227×227 pix. patches are carved from the original images	Same	CA, F1	73.76 – 87.14 (CA), 0.7417 – 0.8728 (F1)	Hand crafted feature descriptors (GLCM and HOG) through a Naïve-Bayes classifier: 68.97 – 82.41 (CA), 0.6931 – 0.8226 (F1)	Yalcin (2017)
18. Keras/Theano	Camera distortion removal, color correction, temporal matching, plant segmentation through the GrabCut algorithm	Image rotations by 90, 180 and 270 degrees around its center	Same	CA	93%	Hand crafted feature descriptors + LSTM: 68% CNN: 76.8%	Namin et al. (2017)
19. MatConvNet	Image segmentation	Simulated roots added to soil images	Same	QM	0.23 (simulation) 0.57 (real roots) 6.298	N/A	Douarre et al. (2016)
20. Caffe	Enhanced Vegetation Index (EVI), hard threshold algorithm, Wavelet transformation for	N/A	Same	RMSE		Support Vector Regression (SVR): 8.204	Kuwata and Shibasaki (2015)

21.	Keras/Theano	detecting crop phenology Intensity image gen., radiometrical calibration, temporal filtering for noise reduction, orthorectification into map coordinates, transformed to logarithm scale, normalized	N/A	Same	CA, F1	99.05% (CA), 0.99 (F1)	RF and SVM (best of both): 91.77% (CA), 0.9179 (F1)	Minh et al. (2017)
22.	TensorFlow	Blurred synthetic images by a Gaussian filter	Generated synthetic 128 × 128 pix. images to train the network, colored circles to simulate background and tomato plant/crops.	Trained entirely on synthetic data and tested on real data	RFC, RMSE	91% (RFC) 1.16 (RMSE) on real images, 93% (RFC) 2.52 (RMSE) on synthetic images	ABT: 66.16% (RFC), 13.56 (RMSE)	Rahneemoonfar and Sheppard (2017)
23.	Caffe	Image segmentation for easier data annotation by users, creation of bounding boxes around image blobs	Training set partitioned into 100 randomly cropped and flipped 320 × 240 pix. sub-images	Same (but different trees used in training and testing)	RFC, L2	0.968 (RFC), 13.8 (L2) for oranges 0.913 (RFC), 10.5 (L2) for apples	Best texture-based regression model: 0.682 (RFC)	Chen et al. (2017)
24.	Caffe	Image segmentation for easier data annotation	Flip, scale, flip-scale and the PCA augmentation technique presented in AlexNet	Same	F1-IoU (20)	0.904 (apples) 0.908 (mango) 0.775 (almonds)	ZF network: 0.892 (apples) 0.876 (mango) 0.726 (almonds)	Bargoti and Underwood (2016)
25.	Caffe	Early/late fusion techniques for combining the classification info from color and NIR imagery, bounding box segmentation, pairwise IoU	N/A	Same (authors demonstrate by using a small dataset that the model can generalize)	F1-IoU (40)	0.838	Conditional Random Field to model color and visual texture features: 0.807	Sa et al. (2016)
26.	Caffe	Resized to 114 × 114 pix., bounding boxes of the object created	Various rotations at 13 scales, intensity of the object adapted	Testing in different fields (row crops, grass mowing), containing other obstacles (people and animals)	CA-IoU (50)	99.9% in row crops and 90.8% in grass mowing	N/A	Steen et al. (2016)

27. Caffe	Image cropping, resized by a factor of 0.75	N/A	Same	F1-IoU (50)	0.72	Local de-correlated channel features: 0.113	Christiansen et al. (2016)
28. Developed by the authors	Image filter extraction through PCA filters bank, binarization and histograms' counting	N/A	Same (also scaling for a certain range translation distance and rotation angle)	CA	90.96%	Manual feature extraction techniques + LMC classifiers: 64.80%	Xinshao and Cheng (2015)
29. Theano-based Lasagne library for Python	Green segmentation to detect green pixels, non-green pixels removal, padding added to make images square, resized to 128×128 pix.	Image mirroring and rotation in 90 degree increments	Same	CA	86.2%	Local shape and color features: 42.5% and 12.2% respectively	Dyrmann et al. (2016a)
30. Caffe	Image cropping	Random flip both horizontally and vertically, random transposing	Same (extra tests for the case of winter barley)	CA	97%	Color feature-based Thistle-Tool: 95%	Sørensen et al. (2017)
31. TensorFlow	Image up-sampling to 299×299 pix., NDVI-based vegetation masks, extracting regions based on a sliding window on the color image	N/A	Same (different carrot fields used for testing)	CA	93.90%	Feature extraction (shape and statistical features) and RF classifier: 85.9%	McCool et al. (2017)
32. Developed by the authors	Resized to 1224×1024 pix.	N/A	Different field used for testing. This field has a severe degree of occlusion compared to the others	IoU	0.64	N/A	Dyrmann et al. (2017)
33. TensorFlow	Separated vegetation/ background based on NDVI, binary mask to describe vegetation, blob segmentation, resized to 64×64 pix., normalized and centered	64 even rotations	Same (also generalized to a second dataset produced 2-weeks after, at a more advanced growth stage)	P, R	Dataset A: 97% (P), 98% (R) Dataset B: 99% (P), 89% (R)	N/A	Milioto et al. (2017)
34. TensorFlow	Pixel-wise segmentation between green vegetation and soil based on NDVI and light CNN, unsupervised dataset summariz.	N/A	Same (also generalized to a second dataset produced 4-weeks after, at a more advanced growth stage)	CA	98% (Dataset A), 59.4% (Dataset B)	Feature extraction (shape and statistical features) and RF classifier: 95%	Potena et al. (2016)

35. Developed by the authors	Image cropping in 800×800 pix.	Random scaling from 80 to 100% of original size, random rotations in one degree increments, varied hue, saturation and intensity, random shadows	Tested on real images while trained on simulated ones	CA, IoU	94% CA, 0.71 IoU (crops), 0.70 IoU (weeds) 0.93 IoU (soil)	N/A	Dyrmann et al. (2016b)
36. Developed by the authors	Geospatial interpolation for creation of soil moisture content maps, multivariate geostatistical approach for estimating thematic soil maps, maps converted to TIFF, resampled to 10-m res.	N/A	Same	RMSE	6.77	Multi-layer perceptron MCA (MLP-MCA): 18% reduction in RMSE	Song et al. (2016)
37. Deep Learning Matlab Toolbox	GLCM features extraction (contrast, energy and homogeneity), saliency maps to accelerate feature extraction	N/A	Same	CA	93.76%	CNN without extra inputs: 89.68% Gaussian Mixture Model (GMM): 90%	Santoni et al. (2015)
38. Developed by the authors	N/A	N/A	Tested on different rooms of pigs than the ones which were used for training	MSE, MRE	0.002 (MSE) on same dataset, 10% (MRE) in relation to a controller	N/A	Demmers et al. (2012)
39. Developed by the authors	N/A	N/A	Tested on different rooms of chicken than the ones which were used for training	MSE, MRE	0.02 (MSE), 1.8% (MRE) in relation to a controller	N/A	Demmers et al. (2010)
40. Keras	N/A	N/A	Same	N-RMSE, MRE	78% (Temperature), 73% (Precipitation), 2.8% (Solar Radiation) N-RMSE, 1–3% MRE in all categories	N/A	Sehgal et al. (2017)

Appendix C. Publicly-available datasets related to agriculture

No.	Organization/Dataset	Description of dataset	Source
1.	Image-Net Dataset	Images of various plants (trees, vegetables, flowers)	http://image-net.org/explore?wnid=n07707451
2.	ImageNet Large Scale Visual Recognition Challenge (ILSVRC)	Images that allow object localization and detection	http://image-net.org/challenges/LSVRC/2017/#det
3.	University of Arcansas, Plants Dataset	Herbicide injury image database	https://plants.uaex.edu/herbicide/ http://www.uaex.edu/yard-garden/resource-library/diseases/
4.	EPFL, Plant Village Dataset	Images of various crops and their diseases	https://www.plantvillage.org/en/crops
5.	Leafsnap Dataset	Leaves from 185 tree species from the Northeastern United States.	http://leafsnap.com/dataset/
6.	LifeCLEF Dataset	Identity, geographic distribution and uses of plants	http://www.imageclef.org/2014/lifeclef/plant
7.	PASCAL Visual Object Classes Dataset	Images of various animals (birds, cats, cows, dogs, horses, sheep etc.)	http://host.robots.ox.ac.uk/pascal/VOC/
8.	Africa Soil Information Service (AFSIS) dataset	Continent-wide digital soil maps for sub-Saharan Africa	http://africasoils.net/services/data/
9.	UC Merced Land Use Dataset	A 21 class land use image dataset	http://vision.ucmerced.edu/datasets/landuse.html
10.	MalayaKew Dataset	Scan-like images of leaves from 44 species classes.	http://web.fsktm.um.edu.my/~cschan/downloads_MKLeaf_dataset.html
11.	Crop/Weed Field Image Dataset	Field images, vegetation segmentation masks and crop/weed plant type annotations.	https://github.com/cwfid/dataset https://pdfs.semanticscholar.org/58a0/9b1351ddb447e6abdede7233a4794d538155.pdf http://www.ipb.uni-bonn.de/data/
12.	University of Bonn Photogrammetry, IGG	Sugar beets dataset for plant classification as well as localization and mapping.	http://flavia.sourceforge.net/
13.	Flavia leaf dataset	Leaf images of 32 plants.	https://www.ideaconnection.com/syngenta-crop-challenge/challenge.php
14.	Syngenta Crop Challenge 2017	2,267 of corn hybrids in 2,122 of locations between 2008 and 2016, together with weather and soil conditions	

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